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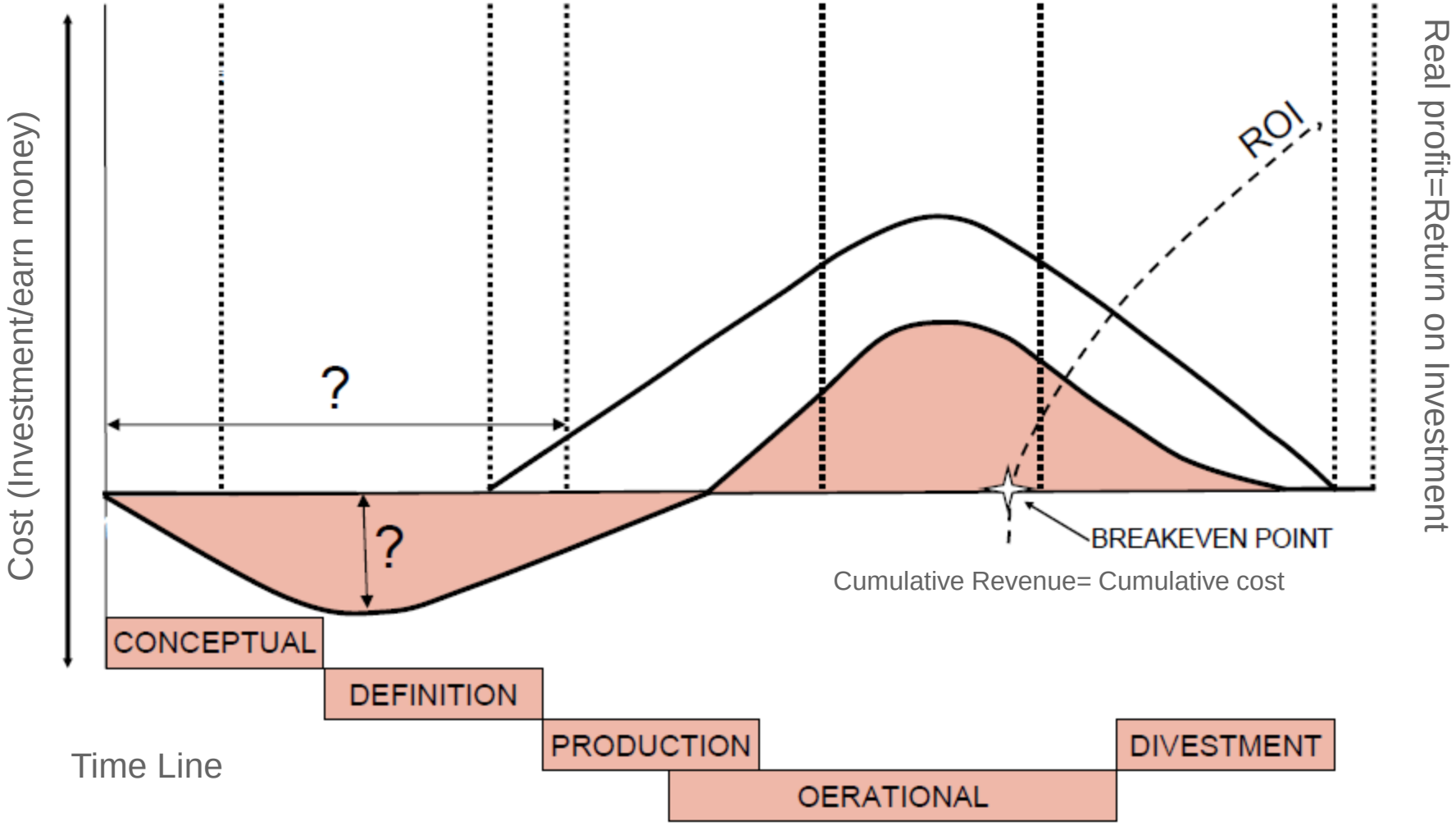
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5. **Process Improvement**
6. Discussion & Summary
7. Bibliography

3 System Product Lifecycle



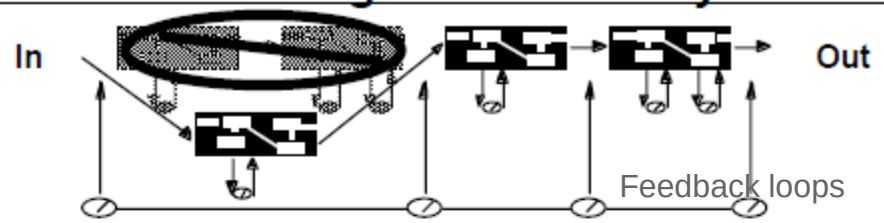
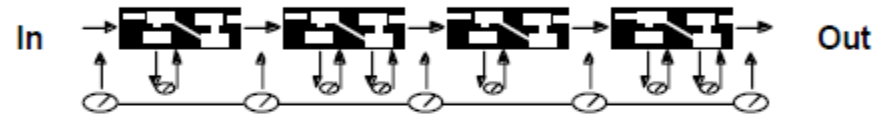
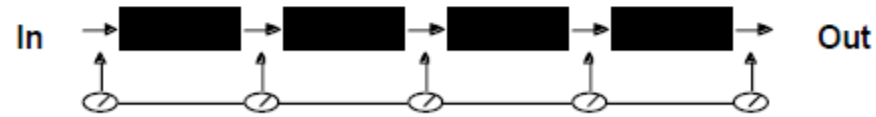
- **Why?**
  - The **quality** outcome and timeliness of the **system** development is highly influenced by the **quality** of the **process** used to acquire, develop, and maintain it. (no luck or talent + well-organized, repeatable process)
- **Common Misconceptions**
  - **I don’t need process, I have**
    - really good people
    - advanced technology
    - an experienced manager
  - **Process**
    - interferes with creativity
    - equals bureaucracy + regimentation
    - isn’t needed when building prototypes
    - is only useful on large projects
    - hinders agility in fast-moving markets
    - costs too much

http://www.sei.cmu.edu/cmmi/general/general.html

- Have you ever been in a project where there was no clear process? What went wrong?

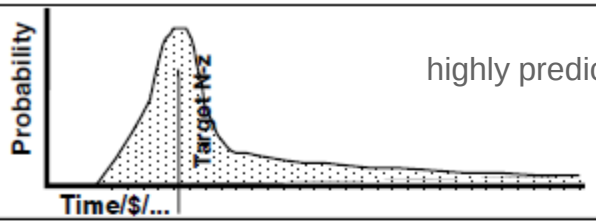
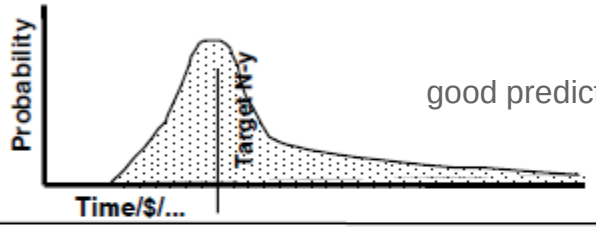
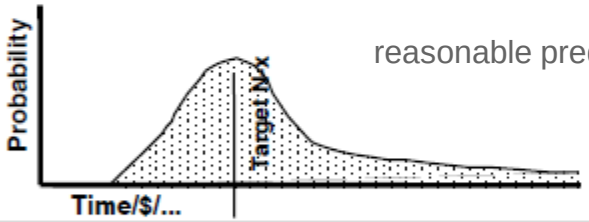
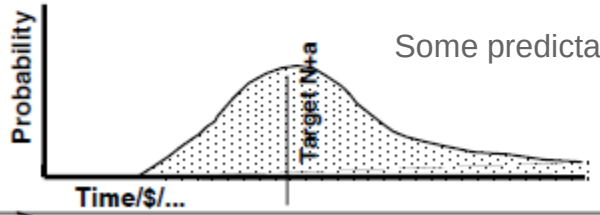
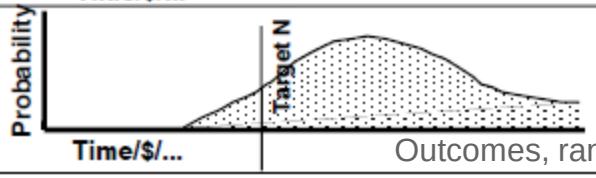
Capability Maturity Model Integration

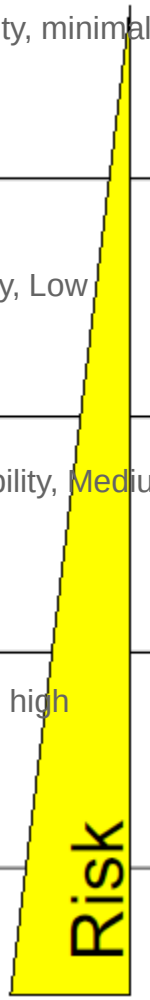
- The CMM Integration Project was formed to define categories **how much a process is supported by a company.**
- **SEI, Carnegie Mellon University)**
- A framework to measure how mature an organization's processes are.
- Not all companies handle their process equally well.
- A visual metaphor for how much management can see what's happening inside the process

Level	Process Characteristics	Management Visibility
5 Optimizing	Focus is on continuous quantitative improvement	
4 Quantitatively Managed	Process is measured and controlled	
3 Defined	Process is characterized for the organization and is proactive	
2 Managed	Process is characterized for projects and is often reactive	
1 Initial	Process is unpredictable, poorly controlled, and reactive	

How predictable the outcome is?

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Level	Process Characteristics	Predicted Performance
5 Optimizing	Focus is on continuous quantitative improvement	 highly predictability, minimal
4 Quantitatively Managed	Process is measured and controlled	 good predictability, Low
3 Defined	Process is characterized for the organization and is proactive	 reasonable predictability, Medium
2 Managed	Process is characterized for projects and is often reactive	 Some predictability, high
1 Initial	Process is unpredictable, poorly controlled, and reactive	 Outcomes, random, very high



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III.6 Discussion & Summary

- We have nearly **the same life cycle models** in the different disciplines.
- Advanced life cycle models and modeldriven approaches try to increase the degree of **automation** and decrease **time to-market**.
- Especially for organizations which develop large-scale software-intensive systems **process improvement** is crucial.

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III.7 Bibliography

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